STAT 17 Week 8
Confidence Intervals

Xiao Zhang

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1. Statistical Inference Overview

Statistical inference allows us to use sample data to estimate population parameters.

Target Parameter

- Quantitative data → population mean μ
- Categorical data → population proportion p

Point Estimate

- $\bar{x}$ estimates μ
- $\hat{p}$ estimates p

Confidence Interval (CI)

$$\text{Point Estimate} \pm (\text{Critical Value})(\text{Standard Error})$$

2. Confidence Level Interpretation

Correct Interpretation:

In the long run, about $(1 - \alpha)\%$ of all possible samples will produce confidence intervals that contain the true parameter.

Important: Confidence is NOT probability.

We do NOT say:

- "There is a 95% probability the parameter is inside."
- "95% of data values fall inside."

3. Confidence Interval for a Population Proportion

When to Use

- Data are Yes/No (success/failure)
- Interested in population proportion p

Conditions

1. Random sample
2. $n \leq 0.05N$
3. $n\hat{p}(1-\hat{p}) \geq 10$

Formula

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Critical values (Z-table):

Confidence Level	$z_{\alpha/2}$
90%	1.645
95%	1.96
99%	2.575

4. Confidence Interval for a Population Mean

Case 1: σ Known (or large n)

When to Use

- Interested in μ
- Population standard deviation σ known
- OR large sample size ($n \geq 30$)

Formula

$$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

Case 2: σ Unknown (Use s)

When to Use

- Interested in μ
- σ unknown (almost always)
- Small sample ($n < 30$)

Conditions

1. Random sample
2. $n \leq 0.05N$
3. Population approximately normal OR $n \geq 30$

Formula

$$\bar{x} \pm t_{\alpha/2,df} \frac{s}{\sqrt{n}}$$

$$df = n - 1$$

5. Z vs T Decision Rule

Parameter	σ Known?	Sample Size	Use
Proportion	N/A	Any	Z
Mean	Yes	Any	Z
Mean	No	$n < 30$	T
Mean	No	$n \geq 30$	Usually T

Key Rule:

- Proportion $\rightarrow$ Always Z
- Mean $\rightarrow$ Check if σ known

6. Margin of Error

$$\text{ME} = (\text{Critical Value})(\text{Standard Error})$$

Effects on Width:

- Increase confidence level $\rightarrow$ wider interval
- Increase sample size $\rightarrow$ narrower interval
- If n quadruples $\rightarrow$ ME is cut in half

7. Sample Size Formulas

For Proportion

$$n = \hat{p}(1 - \hat{p}) \left(\frac{z_{\alpha/2}}{E} \right)^2$$

If no estimate available, use $\hat{p} = 0.5$.

For Mean

$$n = \left(\frac{z_{\alpha/2} s}{E} \right)^2$$

Always round up.

8. Interpretation Templates

Confidence Interval:

We are ___% confident that the true (parameter in context) lies between (lower bound) and (upper bound).

Confidence Level:

In the long run, about ___% of all possible samples will produce intervals that contain the true parameter.

9. Common Exam Traps

- Mixing up parameter vs statistic
- Using Z instead of T for small samples
- Incorrect interpretation of confidence level
- Forgetting to check conditions
- Forgetting degrees of freedom ($df = n - 1$)